SIAM INTERNATIONAL SCHOOL
55/5 Moo.1, Rangsit-Nakhon Nayok Rd. Pathum Thani Thailand
Shaping leaders for tomorrow

Teacher:	Ray Joshua Balingkit	**Subject:**	Computer Science
Week:	11 (Term 1)	**Year:**	10-11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will develop their understanding of condition-controlled repetition and explain how a while loop can execute zero or more times. They will trace a changing condition and identify what allows the loop to terminate. Students will then combine repetition and selection in a text menu, providing useful feedback for an unknown choice and a clear exit option.

End-of-Week Assessment/Check: How will you know they got it?

Teacher questioning, discussion of worked examples, the completed student tasks and the final checks in both lessons. Check termination and explain the zero-iteration case. Test a valid option, an invalid option and quit; check that the menu does not become trapped. Check that students can explain their reasoning and apply the skill without relying on the worked example.

LESSON SLOTS - MONDAY & FRIDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	2 November 2026	By the end of this lesson, students will be able to use a while loop with a condition that changes.	**H:** Predict what happens when a loop counter never changes. Invite students to explain their prediction before demonstrating the example. Use their responses to identify the idea that needs clarification. **ID:** Explain pre-condition testing, initialisation and updating to allow termination. Explain that a while loop tests its condition before the body executes. A loop may therefore run zero times. The program must eventually change the relevant value or provide another clear way to finish. **WD:** Trace a countdown including its initial condition and final value. Start a countdown at 3, display the value and reduce it by 1. Trace the condition before each iteration and after reaching 0. Contrast this	**Support:** Provide a condition-update diagram and a complete countdown with one missing line. Trace values on the whiteboard before coding. Use a stop-and-check routine when students suspect an endless loop. **Extension:** Students change the stopping condition and predict the final displayed value. They explain when a for loop would express the same task more simply and when while is more suitable.

			with starting at 0 and with omitting the update. **YD:** Write a countdown and trace a case where the loop body never runs. Students submit their countdown and traces for a positive start and a zero start. They identify the statement responsible for progress and explain why the condition eventually becomes false. **C:** Check termination and explain the zero-iteration case. Ask students to support their answer with a specific example from their work, and record any misconception that needs a further demonstration.	
2	6 November 2026	By the end of this lesson, students will be able to build a repeating menu with a clear exit option.	**H:** Ask what should happen after an unknown menu choice. Invite students to explain their prediction before demonstrating the example. Use their responses to identify the idea that needs clarification. **ID:** Explain a sentinel choice and selection inside a while loop. Explain that a menu repeats until a sentinel choice is entered and that selection determines the action taken during each repetition. An unknown choice should give feedback and return safely to the menu. **WD:** Model a text menu with one information option and a quit choice. Use choices 1 for information and Q for quit. Trace the sequence 1, X, Q and show when the choice variable is updated. Discuss why reading a new choice only before the loop can prevent the program from exiting.	**Support:** Provide a menu code skeleton with clear places for input, selection and repetition. Use a flow diagram to locate the next input. Model one option and let students complete the remaining branches. **Extension:** Students normalise the quit choice only if they can explain the string operation used, or document case sensitivity as a limitation. They justify their response to an unknown menu option.

			YD: Create a two-action menu that repeats, handles an unknown choice and exits when requested. Students add a second useful text action, keep the menu readable and save the program. A partner tests valid options, an unknown option and quit in one session and records whether each response meets the requirements. **C:** Test a valid option, an invalid option and quit; check that the menu does not become trapped. Ask students to support their answer with a specific example from their work, and record any misconception that needs a further demonstration.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel International GCSE Computer Science Student Book: Chapter 5
- Teacher computer and classroom display
- Classroom computers with Python 3 editor and individual save folders
- Prepared algorithm, tracing or Python practice tasks aligned with the lesson objectives
- Whiteboard and markers
- Student notebooks and pencils
- Success criteria, test tables and teacher observation checklist

END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?